Conditional gradient compression guarantees under a covariance spectral condition

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/ahl-271-compressed-empirical-measures-in-finite-dimensions-2026/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. References inside the span are handed to the assembler as navigation stubs (same-article results the counted proof cites); each stub records non_load_bearing and the reason. Stacks tag stubs are not generated per the S062 chapter config (claim_prefix doi:10.5802/ahl, tag index never consulted).

Editorial note. Printed number of the counted statement in the original: 4.2 (cedram shared theorem counter, section 4); the card reproduces it with initial_counters = {"section": 4, "theo": 1}. The unit's census v3.17 number was drifted for this item; this number was verified against the publisher PDF (ahl.centre-mersenne.org/item/10.5802/ahl.271.pdf) by matching the statement text verbatim.

Editorial note. Citations (none) are neutralized by the no-bibliography card convention.

Editorial note. Judgment note (this lane's acceptance): complete author derivation inside the span; see lane-s11/ahl-271_faces.txt and the deep-read ledger (ahl-271 prop:CGM_CovOpProp, printed 4.2, PDF-verified).

Editorial note. Packaging correction: CGM in the author abstract285 and Algorithm CGM3059 denotes the conditional gradient method for compressing the mean embedding; the previous generated title incorrectly called it conjugate gradient on the covariance operator. Only the generated title and difficulty metadata are corrected; author statement/proof and source are unchanged.

Editorial scope. Single proposition card cut from the AHL article doi:10.5802/ahl.271 (ahl-271); statement and proof are the author's own text at source lines 3092-3114 and 3116-3135.

Allowed standard prerequisites (editorial).

- Covariance-operator theorem (Theorem thm:covariance, printed 2.17). cited inside the counted statement/proof spans. the first statement
- Kernel-herding guarantees (Proposition prop:KH_guarantees, printed 4.1). cited inside the counted statement/proof spans. the argument reused in the support-restricted space

Source reference. Original theorem 2.17: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/ahl-271-compressed-empirical-measures-in-finite-dimensions-2026/pinned-original-source.tex, source lines 1773–1773. navigation stub only: the target lives outside the counted span in the same article; the counted proof reads the cited result/identity from the stub text, so the stub is not load-bearing

Source reference. Original proposition 4.1: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/ahl-271-compressed-empirical-measures-in-finite-dimensions-2026/pinned-original-source.tex, source lines 2903–2903. navigation stub only: the target lives outside the counted span in the same article; the counted proof reads the cited result/identity from the stub text, so the stub is not load-bearing

Proposition 4.2. *Let $(\mathcal{X}, \mathcal{A}, P)$ be some probability space with P being a topological measure that is τ-additive, and with measurable kernel function k defined on $\mathcal{X}$ such that the corresponding RKHS $\mathcal{H}$ is finite dimensional. Furthermore, let $X_1, \ldots, X_n$ be i.i.d. random variables attaining values in $\mathcal{X}$ and with law P. Assume that $\|k\|_\infty < \infty$, and that the centered covariance operator $\widetilde{\mathfrak{C}}_c$ has an eigen-decomposition with smallest non-zero eigenvalue being $\bar{\lambda}_d$. Let $\beta = 3\|k\|_\infty^{1/2}/\delta$ then with probability $1-q$, $q \in (0,1)$,*

$$\left\|\widehat{\mathfrak{m}}_{\lceil 2\beta \log(n)\rceil} - \mathfrak{m}_n\right\| \leq 2\|k\|_\infty^{1/2} n^{-1/2}$$

whenever n is (strictly) greater than

$$\left(\frac{8\|k\|_\infty\left(\sqrt{2\log(4/q)} + 192\|k\|_\infty/\bar{\lambda}_d\right)}{\bar{\lambda}_d^2}\right)^2 \vee \left(\frac{12\|k\|_\infty\left(1+\sqrt{\log(2/q)}\right)}{\bar{\lambda}_d}\right)^2$$

Let $\delta = \bar{\lambda}_d/2\|k\|_\infty^{1/2}$. *For any* $t \geq 1, n \geq 1$,

$$E(\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\|) \leq \exp\Big(-\delta(t-1)/24\|k\|_\infty^{1/2}\Big) + 6\|k\|_\infty^{1/2}\exp\big(-(c_3 n^{1/2} - c_4)_+^2\big),$$

where $c_3 = (\bar{\lambda}_d^2/32\|k\|_\infty) \wedge (\bar{\lambda}_d/12\|k\|_\infty)$ *and* $c_4 = (48\|k\|_\infty/\bar{\lambda}_d) \vee 1$ *are possible choices.*

If the stopping criterion of the algorithm is an error of $\|\widehat{\mathfrak{m}}_{\hat{t}} - \mathfrak{m}_n\| \leq n^{-1/2}$, *i.e.* $\widehat{t} = \inf_{t\geq 1}\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq n^{-1/2}$, *and if the infimum is greater than* n, *then* $\widehat{t} = n$ *and* $\widehat{\mathfrak{m}}_{\hat{t}} = \mathfrak{m}_n$, *then*

$$E(\widehat{t}) \leq \Big\lceil 1 + 12\|k\|_\infty^{1/2}\log(n)/\delta \Big\rceil + 3n\exp\big(-(c_3 n^{1/2} - c_4)_+^2\big).$$

Proof. The *first statement* follows directly from Theorem 2.17 and the bound on the error of the CGM. For the *other statements* let us consider the space $\mathcal{H}_S$ corresponding to the kernel $k_S = k\restriction S \times S$, where S is the support of the law P, and with corresponding objects $\mathfrak{m}_S, \mathfrak{m}_{S,n}$ and $\widetilde{\mathfrak{C}}_c^S$. As in the proof of Proposition 4.1, we have that

$$\begin{aligned}\Pr\big(\|\mathfrak{m}_{S,n} - \mathfrak{m}_S\|_S \geq \delta/4\big)& \\ \leq \exp\left(-\left(\frac{\delta n^{1/2}}{\sqrt{32}\|k\|_\infty^{1/2}} - 1\right)^2\right) &\leq \exp\left(-\left(\frac{n^{1/2}\bar{\lambda}_d}{12\|k\|_\infty} - 1\right)^2\right)\end{aligned}$$

whenever $n \geq (\sqrt{32}\|k\|_\infty^{1/2}/\delta)^2$. Furthermore, there is a ball or radius $\delta/2$ around $\mathfrak{m}_S$ in C_n (as a subset of the affine span of C_S) with probability at least

$$1 - 2\exp\left(-\left(\frac{n^{1/2}\bar{\lambda}_d^2}{32\|k\|_\infty} - \frac{48\|k\|_\infty}{\bar{\lambda}_d}\right)^2\right),$$

for large enough n. Hence, with probability at least

$$1 - 3\exp\big(-(c_3 n^{1/2} - c_4)_+^2\big)$$

there is a ball or radius $\delta/4$ around $\mathfrak{m}_{S,n}$ in C_n (as a subset of the affine span of C_S). The second result follows since the CGM reduces the error in each step and the initial error is bounded by $\|\widehat{\mathfrak{m}}_1 - \mathfrak{m}_n\| \leq 2\|k\|_\infty^{1/2}$.

The *third statement* follows along similar lines. In the event that we have a ball of size $\delta/4$ it follows that $\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq \exp(-\delta(t-1)/24\|k\|_\infty^{1/2})$. Setting the right hand side to $n^{-1/2}$ leads to $\widehat{t} \leq \lceil 1 + 12\|k\|_\infty^{1/2}\log(n)/\delta\rceil$. □